Grafana Monitoring AWS EC2 Instance Metrics: Complete Guide & Step-by-Step

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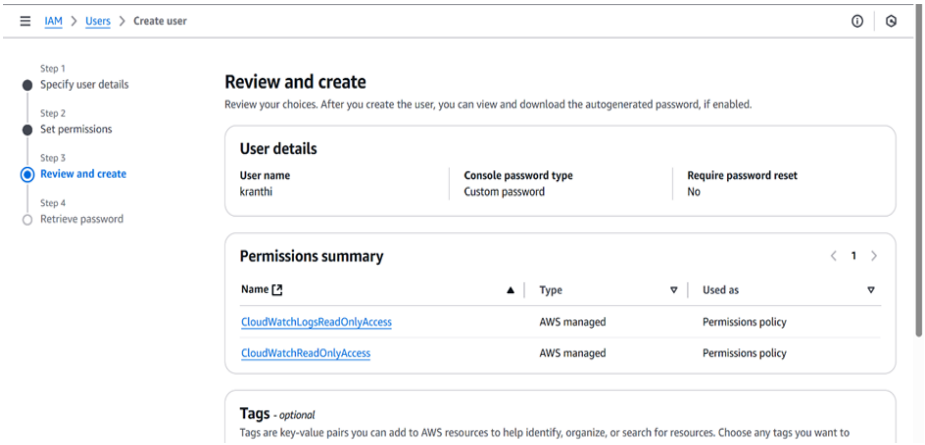
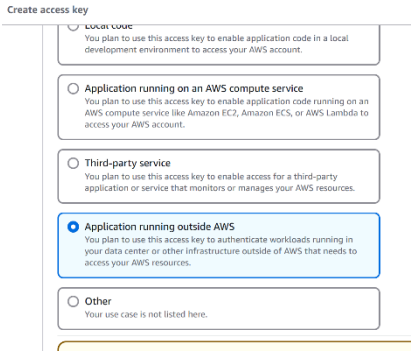
Grafana Monitoring for AWS EC2 Server

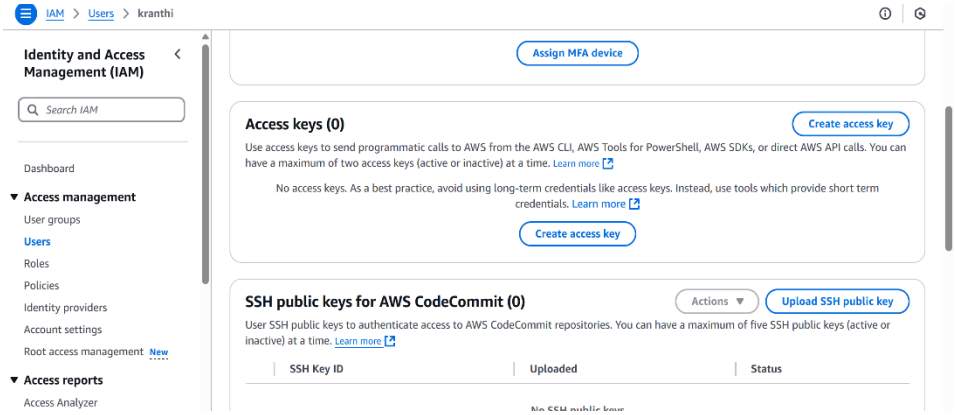
This document provides a step-by-step guide to setting up **Grafana** for monitoring an AWS EC2 instance. It covers installation, configuration, and visualization of system metrics. The guide ensures real-time monitoring of CPU, memory, disk usage, and network activity to optimize server performance and detect issues proactively.

Step 1: Create an IAM User for Grafana

- Navigate to **AWS IAM Console** → **Users** → **Create User**.
- Assign the following AWS-managed policies:
    - **CloudWatchReadOnlyAccess**
    - **CloudWatchLogsReadOnlyAccess**
- Generate **Access Key** and **Secret Access Key**
    - Store the credentials securely







Step 2: Install Grafana

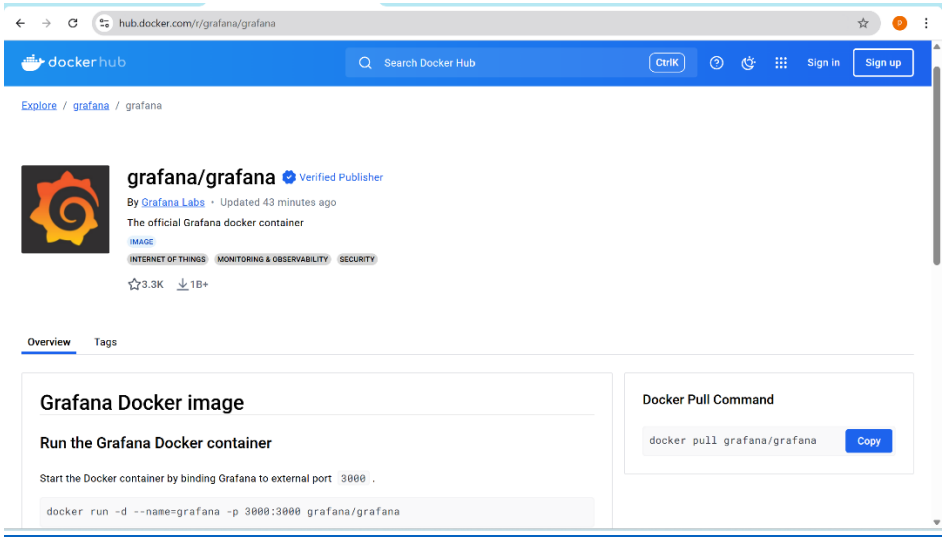
Choose one of the installation methods:

Official Package – Install using the official repository https://grafana.com/docs/grafana/latest/setup-grafana/installation/

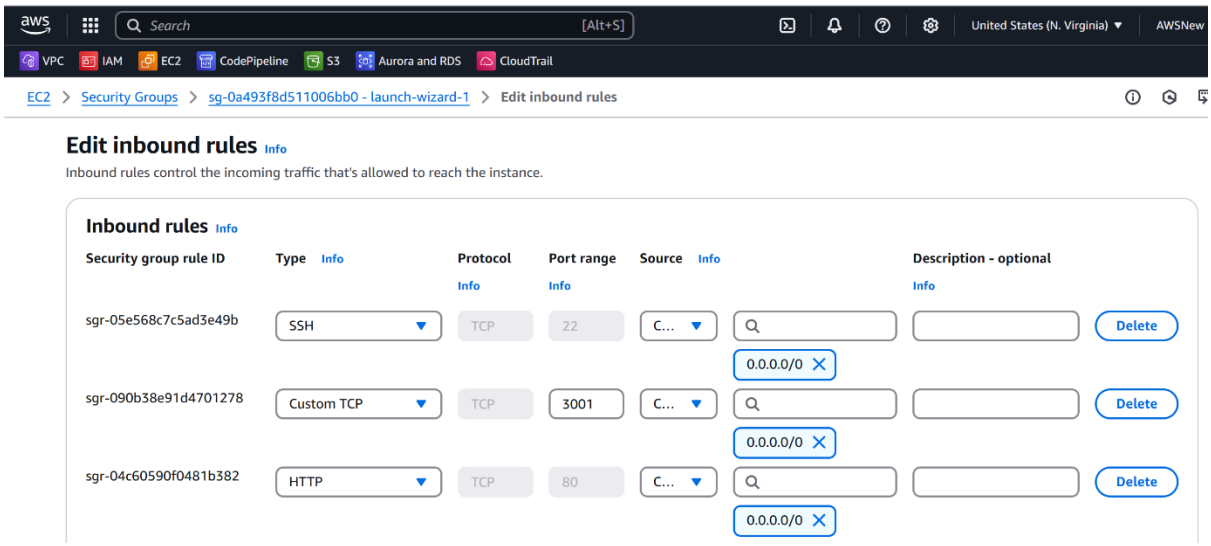
Grafana Cloud – Sign up at Grafana Cloud https://grafana.com/products/cloud/

Docker – Deploy Grafana using: docker run -d -p 3000:3000 --name=grafana grafana/grafana

https://hub.docker.com/r/grafana/grafana

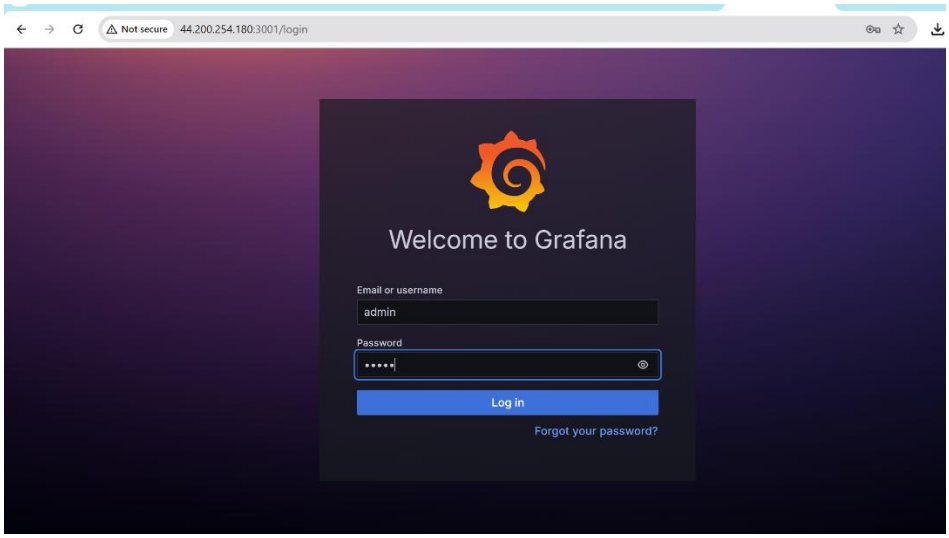


- Allow **TCP port 3000** in the EC2 security group for Grafana access.



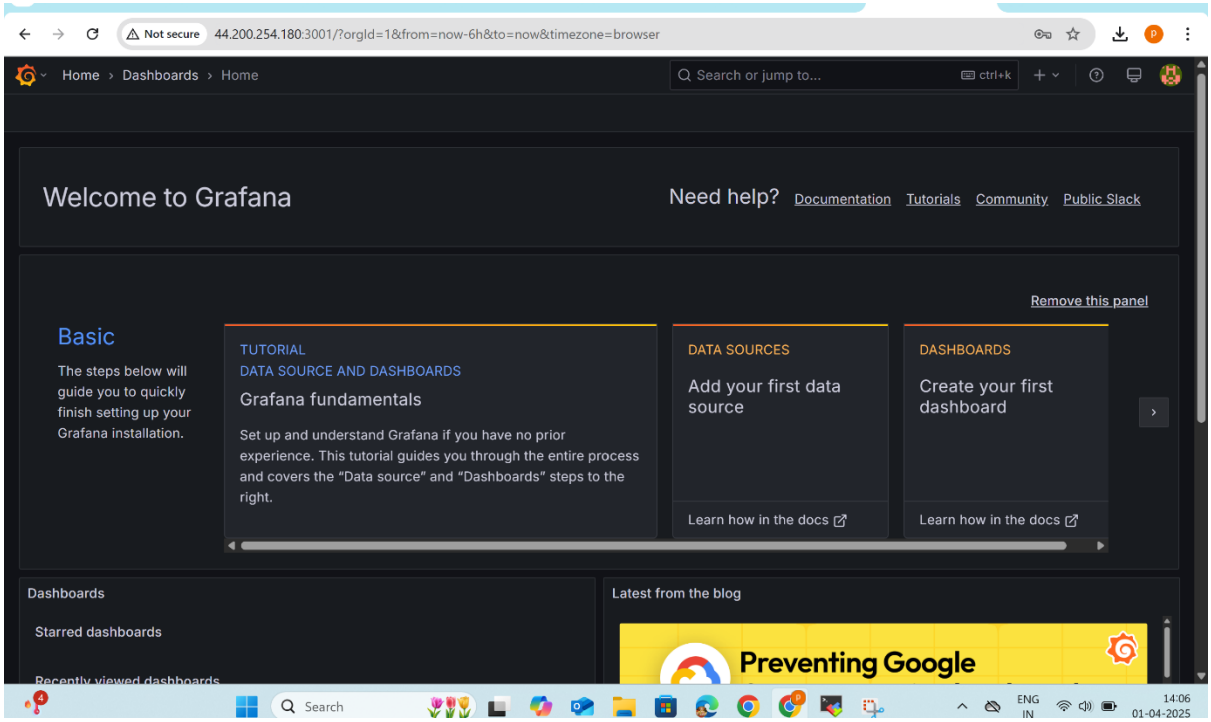
Step 3: Sign in to Grafana

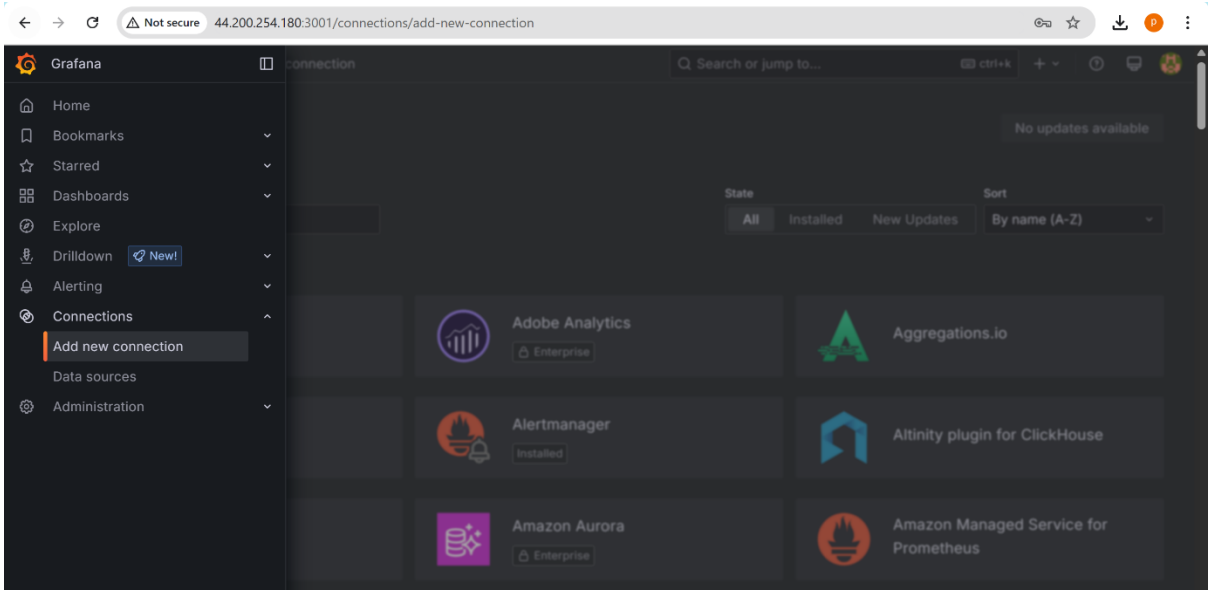
Log in with **Username: admin** and **Password: admin**, then set a new custom password.



Step 4: Add Data Source

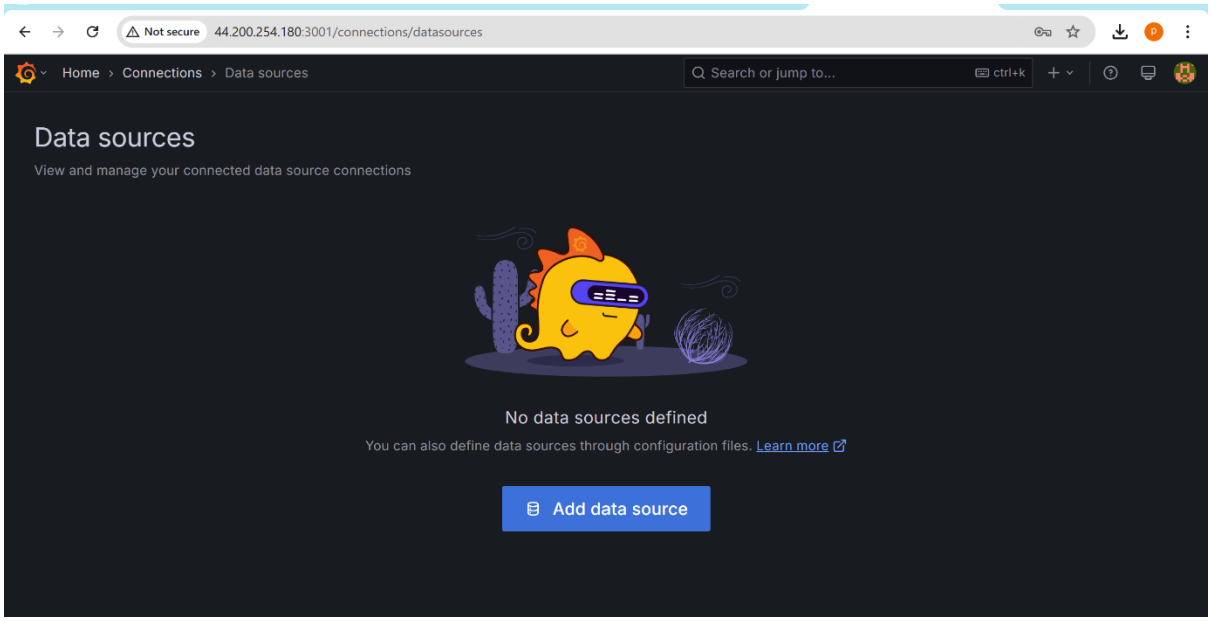
Go to **Connections → Data Sources**, select **CloudWatch**, and configure it for your AWS EC2 server.

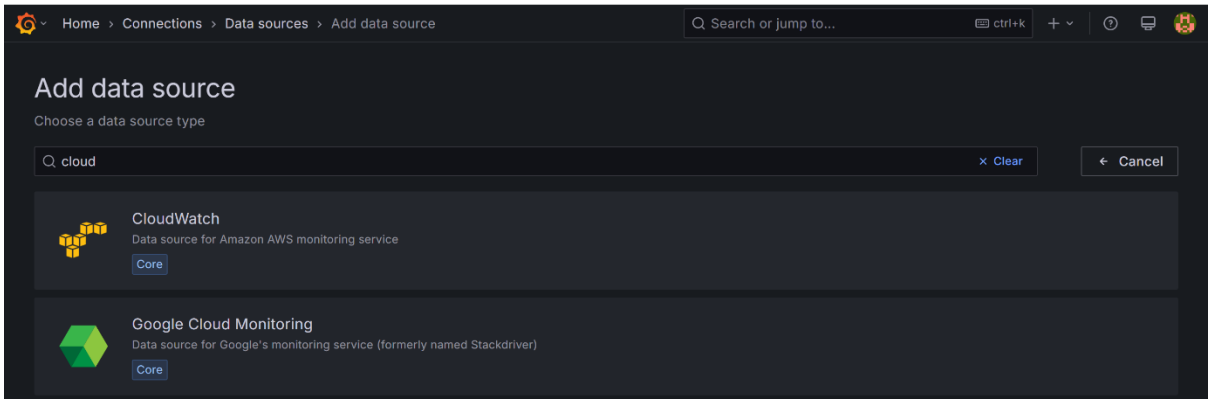




Step 5: Select CloudWatch as Data Source

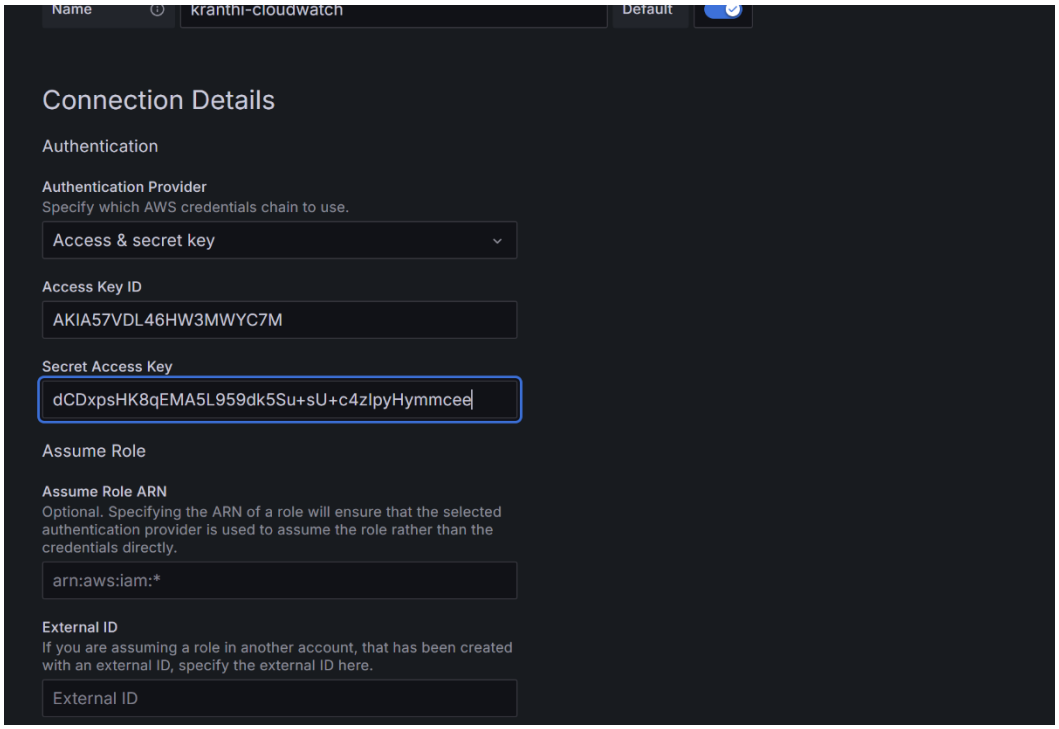
Click **Add data source**, search for **CloudWatch**, and select it.



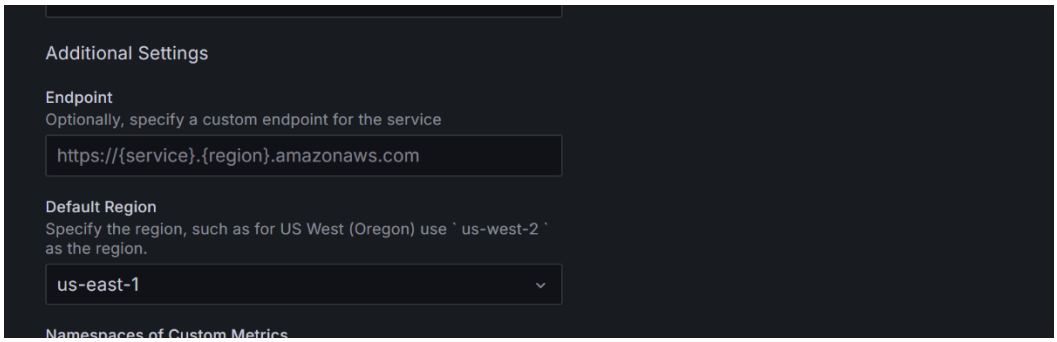


Step 6: Configure CloudWatch Connection

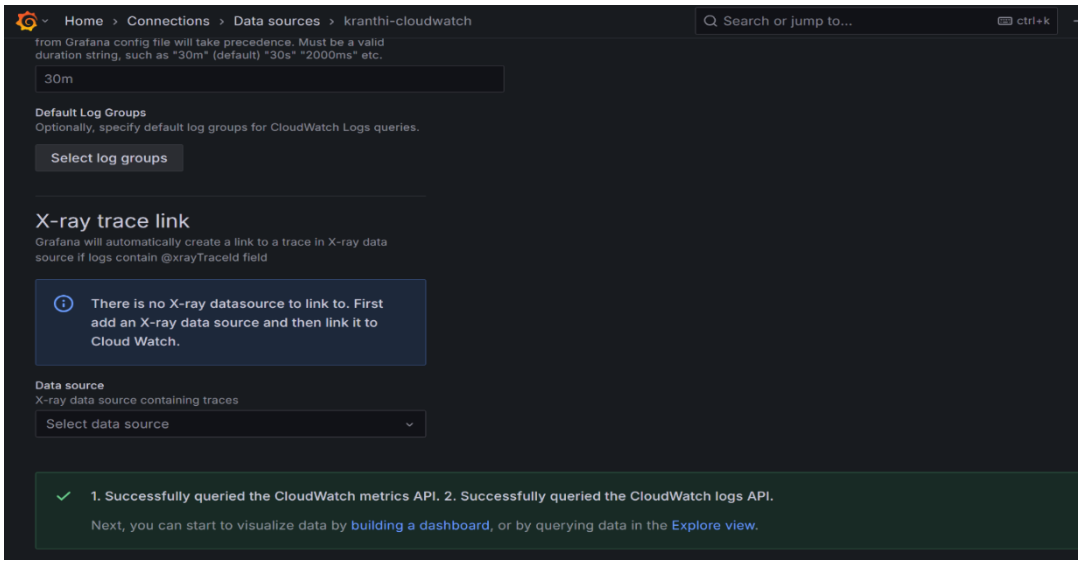
- Enter **Access Key** and **Secret Access Key**.



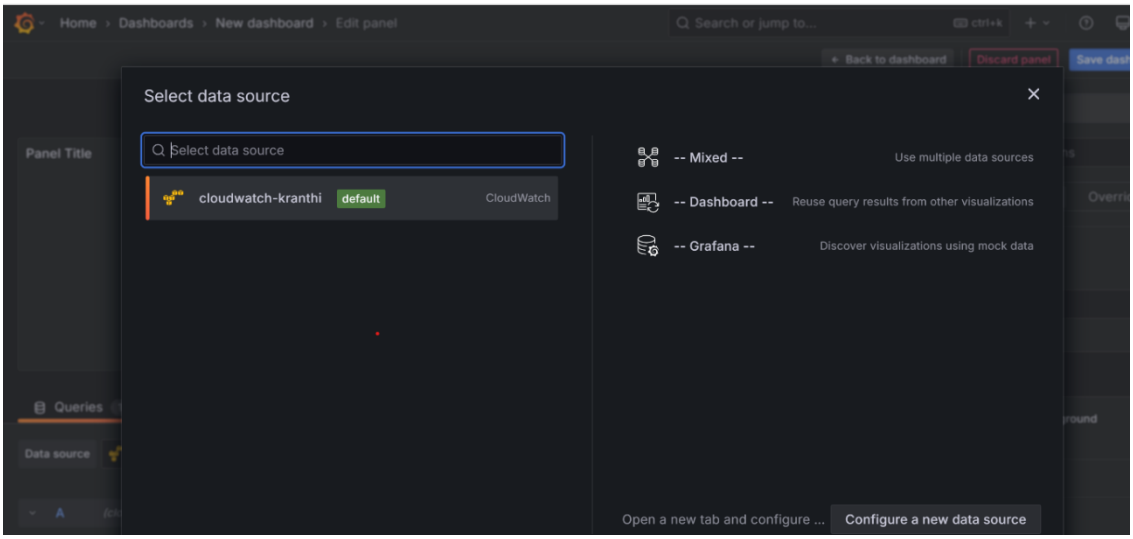
- Select **AWS Region** and set **Namespace** to AWS/EC2.



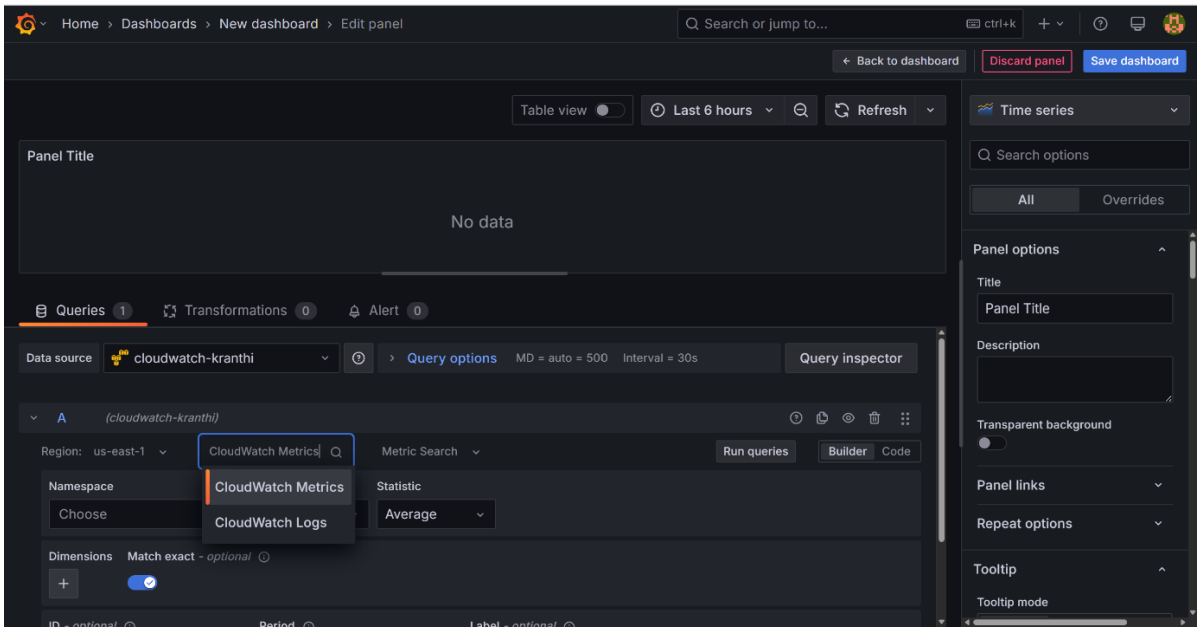
- Click **Test** to verify the connection (should show **Success**).

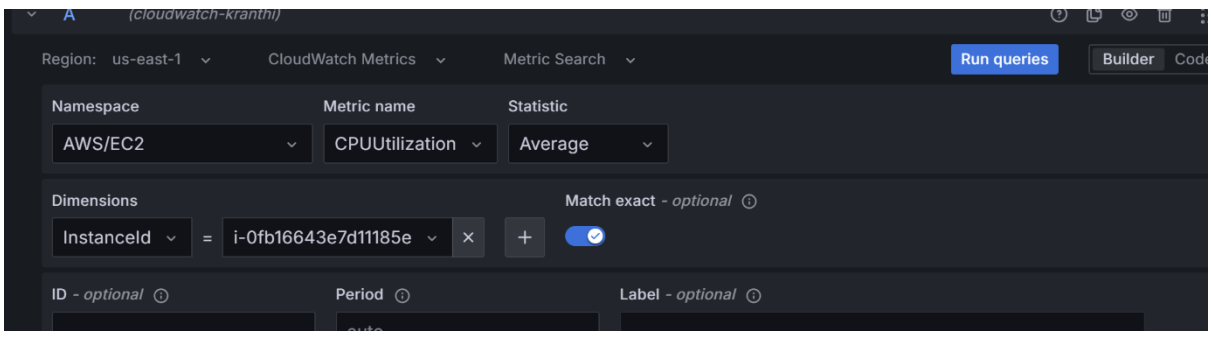


- Go to **Home → Dashboards → New Dashboard**.
- Select **Add Query** and choose the **CloudWatch data source**.



- Set **Region**, **Namespace (AWS/EC2)**, and select metrics (CPU, Storage, Networking, IOPS, etc.).
- Click **Run Query** to fetch data.





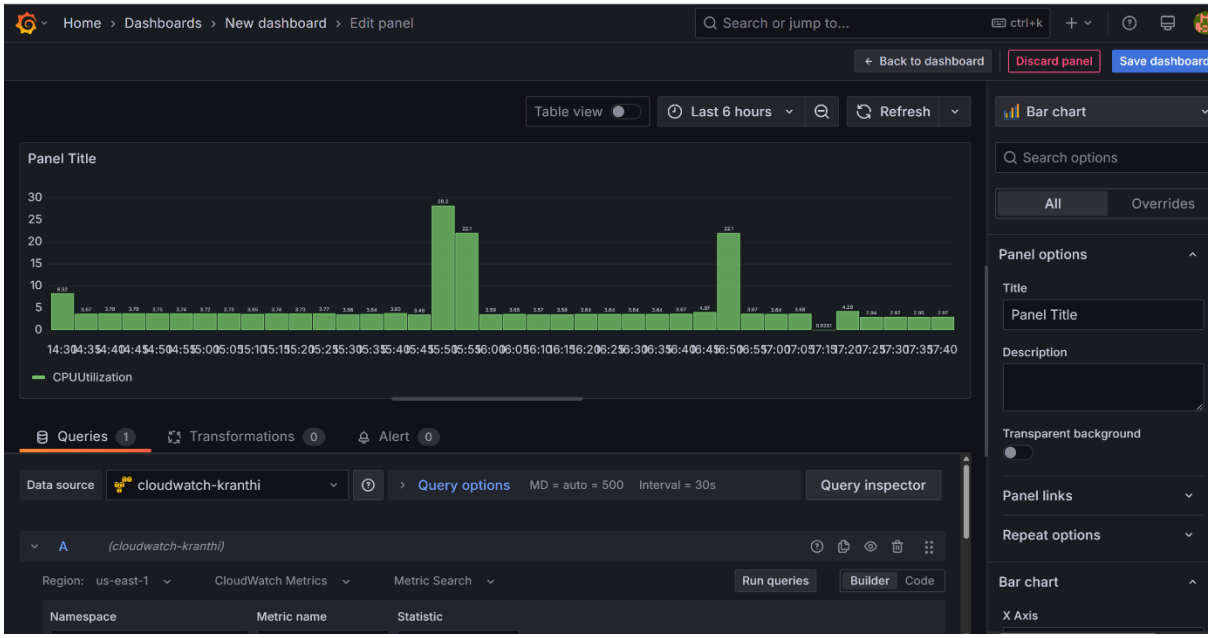
- To see spikes in visualization, run a stress test on EC2: stress –cpu 4 –timeout 60

```
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :
  Installing       : stress-1.0.7-2.amzn2023.0.1.x86_64
  Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64
  Verifying        : stress-1.0.7-2.amzn2023.0.1.x86_64

Installed:
  stress-1.0.7-2.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-9-50 ~]$ stress --cpu 4 --timeout 60
stress: info: [28198] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
stress: info: [28198] successful run completed in 60s
[ec2-user@ip-172-31-9-50 ~]$ stress --cpu 4 --timeout 60
stress: info: [28322] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
stress: info: [28322] successful run completed in 60s
[ec2-user@ip-172-31-9-50 ~]$
```

- Visualize Data and you can change visualization by using Bar charts



Conclusion

Grafana provides powerful monitoring and visualization for AWS EC2. Beyond EC2 metrics, you can:

- Monitor **RDS, S3, Lambda, and other AWS services** via CloudWatch.
- Set up **alerts** for resource spikes or failures.
- Integrate with **Prometheus, Loki (logs), and New Relic** for deeper insights.
- Create custom **dashboards** for multi-cloud and hybrid environments.

Grafana helps in proactive monitoring, ensuring better performance and uptime.

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